Audio Representation Learning and Downstream Task Modeling: A Survey

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A decoupled framework for audio foundation models

Task-specific views obscure how reusable audio features enter downstream systems. We separate **audio-to-feature representation learning** from **feature-to-target downstream modeling**. The survey reviews modeling domains, representation forms, architectures, and pre-training; categorizes **five integration mechanisms**; and connects **discriminative, autoregressive, non-autoregressive, and hybrid modeling** to representative systems. This framework guides choices among temporal resolution, acoustic fidelity, semantic alignment, and latency.

01 Audio representation learning: what to preserve

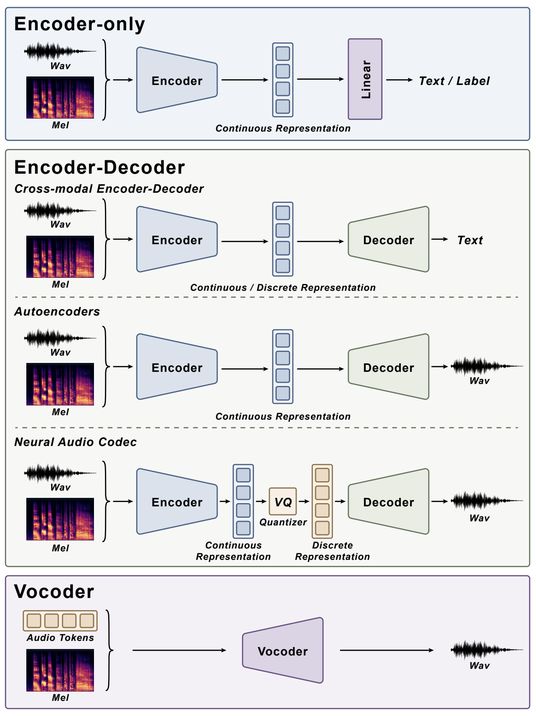


Figure 1. Representation architectures and waveform reconstruction.

Modeling domain

Time domain. Raw waveforms preserve phase and transients for reconstruction and low-latency applications; long sequences increase computational cost.

Time-frequency domain. Spectrograms shorten sequences and expose harmonic structure for understanding, while imposing time-frequency resolution and phase-modeling constraints.

Representation form

Continuous features. Dense acoustic detail supports understanding, diffusion, and flow matching. Adapters usually map these features into language-model spaces.

Discrete tokens. Symbols unify audio-text sequence modeling but incur quantization loss. Acoustic tokens support reconstruction; semantic tokens emphasize phonetic or linguistic content.

Architectures and components

Encoder-only. Extract discriminative features for lightweight prediction heads or cross-modal retrieval.

Cross-modal encoder-decoders. Map latent features into a different modality, as in audio-to-text speech recognition.

Autoencoders. Learn compressed features through reconstruction. Lightweight pre-training decoders may be discarded after training.

Neural audio codecs. Add quantization to compress waveforms into discrete tokens, then reconstruct audio with a generative decoder.

Vocoders. Reconstruct waveforms from acoustic intermediates such as Mel-spectrograms or codec tokens.

Pre-training paradigms

Supervised pre-training learns from labels or paired targets; annotation cost limits scale. **Self-supervised learning** uses audio-derived targets: **predictive** learning infers masked content or teacher features, **contrastive** learning aligns related examples and separates unrelated ones, and **reconstructive** learning restores missing or compressed signals.

02 Five integration mechanisms: how features enter the model

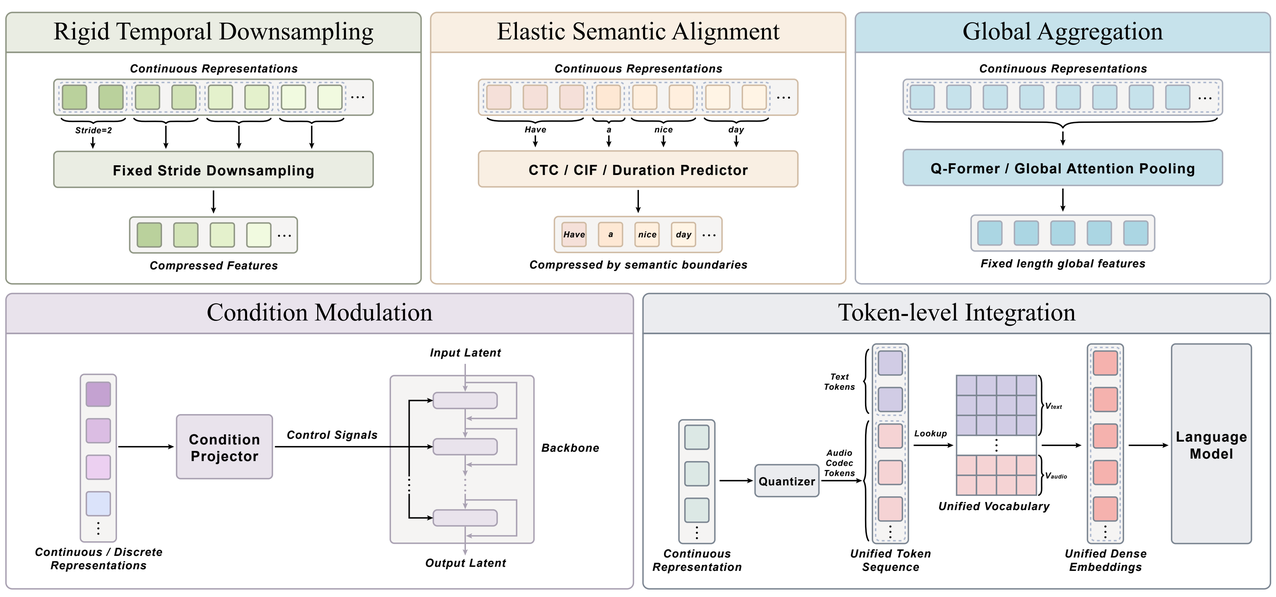


Figure 2. Five ways to inject audio representations into a downstream backbone.

RTD | Rigid temporal downsampling
Fixed-stride pooling or convolution compresses sequences while preserving order. It reduces computation but can smooth away fine acoustic detail.

ESA | Elastic semantic alignment
CTC, CIF, or duration-based mechanisms adapt compression to semantic boundaries, aligning uneven acoustic sequences with linguistic units.

GA | Global aggregation
Queries or attention pooling produce fixed-size global summaries for classification and retrieval, at the cost of fine temporal detail.

CM | Condition modulation
FiLM, AdaLN, or cross-attention uses representations to control backbone activations, injecting global style or sequence conditions into generation.

TI | Token-level integration
Quantized symbols enter dedicated or expanded vocabularies for unified text-audio sequence learning and dialogue, with a quantization bottleneck.

03 Downstream task modeling: how to predict the target

Discriminative
Map features directly to labels or continuous targets for classification, enhancement masks, and source separation.

Autoregressive
Generate causal sequences conditioned on earlier outputs. Coherent text-audio interaction comes with the latency of sequential prediction.

Iterative NAR
Refine a sequence in parallel using global context. Diffusion and flow matching trade refinement steps for acoustic fidelity.

One-shot NAR
Predict a sequence in one forward pass. Alignment or distilled mappings support efficient synthesis with no iterative refinement.

Hybrid
Combine AR semantic planning with NAR acoustic refinement. Discriminative predictions can also guide generation.

04 Representative systems and inference pipelines

Table 1 summarizes representative audio systems, relating their representations, integration mechanisms, and modeling paradigms to inference pipelines and reported benchmark performance.

Cat.	Task	Benchmark & Metrics	Model	Downstream Inference Data Flow Pipeline	Performance
Und	ASR	**LibriSpeech** test-clean WER↓|test-other WER↓	CJST (Zhou et al., 2025)	$[\mathcal{M}\to\text{VGG}^{\text{RTD}}\to\text{Conformer}^{\text{ESA}}\to H_{\text{A}}]\to\text{Dec-only Tr}^{\text{AR}}\to\mathcal{T}$	2.09% | 4.71%
			MDM-ASR (Yen et al., 2026)	$[\mathcal{M}\to\text{FastConformer}^{\text{RTD}}\to H_{\text{A}}^{\text{CM}}]\ \Vert\ [Z_{\text{T}}]\to\text{Non-causal Tr Dec}^{\text{iNAR}}\to\mathcal{T}$	1.80% | 3.60%
			Qwen3-Omni-30B-A3B-Instruct (Qwen Team, 2025)	$[\mathcal{M}\to\text{AuT}^{\text{RTD}}\to H_{\text{A}}]\oplus[Z_{\text{T}}]\to\text{Qwen3}^{\text{AR}}\to\mathcal{T}$	**1.22%** | **2.48%**
	AQA	**MMAR** ACC↑	DiFFA-2 (Zhou et al., 2026b)	$[\mathcal{M}\to\text{Whisper Encoder}\to H_{\text{A}}\to[\text{CNN}\to\text{Linear}^{\text{RTD}}\to H_{\text{A}}]\oplus[\text{Q-Former}^{\text{GA}}\to H_{\text{A}}]]\ \Vert\ [Z_{\text{T}}]\to\text{LLaDA}^{\text{iNAR}}\to\mathcal{T}$	50.80%
			AF-Next-Think (Ghosh et al., 2026a)	$[\mathcal{M}\to\text{AF-Whisper Encoder}\to H_{\text{A}}\to\text{Stride-2 Pool}^{\text{RTD}}\to\text{MLP}\to H_{\text{A}}]\oplus[\mathcal{T}]\to\text{Qwen2.5}^{\text{AR}}\to\mathcal{T}$	61.00%
			Audio-DeepThinker (He et al., 2026)	$[\mathcal{M}]\ \Vert\ [\mathcal{T}]\to\text{Qwen3-Omni-30b-A3B}^{\text{AR}}\to\mathcal{T}$	**74.00%**
Gen	TTS	**Seed-TTS-Eval** test-zh CER↓|test-zh SIM↑| test-en WER↓|test-en SIM↑	OmniVoice (Zhu et al., 2026)	$[\mathcal{W}\to\text{Higgs-Audio Tok}\to Z_A^{\text{TI}}]\oplus[\mathcal{T}]\to\text{Qwen3}^{\text{iNAR}}\to Z_{\text{A}}\to\text{Higgs-Audio Detok}\to\mathcal{W}$	1.60% | 0.741 | **0.84%** | **0.777**
			CosyVoice 3 (Du et al., 2025)	$[\mathcal{W}\to\text{FSQ-MinMo}\to Z_A^{\text{TI}}]\oplus[\mathcal{T}]\to\text{Tr}^{\text{AR}}\to Z_{\text{A}}^{\text{CM}}\to\text{DiT}^{\text{iNAR}}\to\mathcal{M}\to\text{Vocoder}\to\mathcal{W}$	**0.71%** | **0.775** | 1.45% | 0.695
			VibeVoice (Peng et al., 2026b)	$[\mathcal{W}\to\text{Acoustic Tok}^{\text{RTD}}\to H_{\text{A}}]\oplus[\mathcal{T}]\to\text{Qwen2.5}^{\text{AR}}\to H_{\text{A}}^{\text{CM}}\to\text{Diffusion}^{\text{iNAR}}\to H_{\text{A}}\to\text{Acoustic Dec}\to\mathcal{W}$	1.16% | 0.744 | 3.04% | 0.689
	TTA	**AudioCaps** FD_{o3} ↓| KL_p ↓| FD ↓| FAD ↓| KL ↓| CLAP ↑	TangoFlux (Hung et al., 2026)	$[\mathcal{W}\to\text{Stable Audio Open VAE}\to H_{\text{A}}^{\text{CM}}]\oplus[\mathcal{T}\to\text{FLAN-T5}\to H_{\text{T}}^{\text{CM}}]\to\text{DiT}^{\text{iNAR}}\to H_{\text{A}}\to\text{Stable Audio Open VAE Dec}\to\mathcal{W}$	83.58 | 0.95 | 1.57 | 2.34 | 0.52 | **0.385**
			AudioLDM 2 (Liu et al., 2024b)	$\mathcal{T}\to[\text{CLAP}]\ \Vert\ [\text{FLAN-T5}]\to H_{\text{T}}\to\text{GPT-2}^{\text{AR}}\to H_{\text{T}}^{\text{CM}}\to\text{Transformer-UNet}^{\text{iNAR}}\to H_{\text{A}}\to\text{VAE Dec}\to\mathcal{M}\to\text{HiFi-GAN}\to\mathcal{W}$	87.74 | 1.01 | 1.59 | 2.63 | 0.57 | 0.252
			AudioStory (Guo et al., 2025)	$[\mathcal{M}\to\text{Whisper Encoder}^{\text{RTD}}\to H_{\text{A}}]\oplus[\mathcal{T}_p]\to\text{Qwen2.5}^{\text{AR}}\to[\mathcal{T}\ \Vert\ Z_{\text{A}}^{\text{CM}}]\to\text{TangoFlux}^{\text{iNAR}}\to H_{\text{A}}\to\text{VAE Dec}\to\mathcal{W}$	**83.39** | **0.91** | **1.52** | **2.29** | **0.51** | 0.383
Conv	VC	**Seed-TTS-Eval** test-zh CER↓|test-zh SIM↑| test-en WER↓|test-en SIM↑	MeanVC (Ma et al., 2026a)	$[\mathcal{W}\to\text{ECAPA-TDNN}^{\text{GA}}\to H_{\text{A}}^{\text{CM}}]\oplus[[[\mathcal{W}\to\text{Fast-U2++}\to H_{\text{A}}]\ \Vert\ [\mathcal{M}]]\to\text{Timbre Enc}\to H_{\text{A}}^{\text{CM}}]\to\text{DiT}^{\text{1-NAR}}\to\mathcal{M}\to\text{Vocos}\to\mathcal{W}$	3.89 | **0.73** | — | —
			X-VC (Zheng et al., 2026)	$[\mathcal{W}\to\text{SAC Enc}^{\text{RTD}}\to H_{\text{A}}]\ \Vert\ [[\mathcal{W}\to\text{ERes2Net}\to H_{\text{A}}^{\text{CM}}]\oplus[\mathcal{M}^{\text{CM}}]]\to\text{Tr}^{\text{1-NAR}}\to H_{\text{A}}\to\text{SAC Dec}\to\mathcal{W}$	1.99 | **0.73** | 2.83 | **0.63**
			Seed-VC tiny (Liu, 2024)	$[\mathcal{M}\to\text{Whisper Encoder}\to H_{\text{A}}^{\text{CM}}]\oplus[\mathcal{W}\to\text{CAM++}\to H_{\text{A}}^{\text{CM}}]\oplus[\mathcal{W}\to\text{RMVPE}\to H_{\text{A}}^{\text{CM}}]\to\text{U-DiT}^{\text{iNAR}}\to\mathcal{M}\to\text{BigVGANv2}\to\mathcal{W}$	**1.79** | 0.60 | **2.24** | 0.41
Int	FSD	**Full-Duplex-Bench** DMOS↓ | SSIM↑ | Lat↓ | GPT4o↑	PersonaPlex (Roy et al., 2026)	$\mathcal{W}\to\text{Moshi}^{\text{AR}}\to\mathcal{W}$	3.90 | **0.57** | **0.070** | **4.210**
			Moshi (Défossez et al., 2024)	$[\mathcal{W}\to\text{Mimi Tok}^{\text{RTD}}\to Z_{\text{A}}^{\text{TI}}]\to\text{RQ-Tr}^{\text{AR}}\to[Z_{\text{T}}\ \Vert\ Z_{\text{A}}]\to\text{Mimi Detok}\to\mathcal{W}$	**3.11** | 0.10 | 0.265 | 0.765
			Freeze-Omni (Wang et al., 2025c)	$[\mathcal{M}\to\text{Speech Enc}\to\text{Adapter}^{\text{RTD}}\to H_{\text{A}}]\to\text{Qwen2-7B}^{\text{AR}}\to[H_{\text{A}}\ \Vert\ Z_{\text{T}}]\to\text{Speech Dec}^{\text{AR}\ \text{1-NAR}}\to Z_{\text{A}}\to\text{TiCodec}\to\mathcal{W}$	3.51 | 0.05 | 0.953 | 3.615

Tasks: ASR = speech recognition; AQA = audio question answering; TTS = text-to-speech; TTA = text-to-audio; VC = voice conversion; FSD = full-duplex spoken dialogue.
Notation: W / M / T = waveform / Mel / text; H_{A} / H_{T} = continuous audio / text features; Z_{A} / Z_{T} = discrete tokens. Double bar = parallel branches; circled plus = fusion.
Generation: AR = autoregressive; NAR = non-autoregressive; iNAR = iterative NAR; 1-NAR = one-shot NAR. **Components:** Tr = Transformer; Enc / Dec = encoder / decoder; Tok / Detok = tokenizer / detokenizer.
Results: bold / underline = best / second best. Scales, training data, and protocols differ; these are not controlled comparisons. Survey coverage: early 2026.